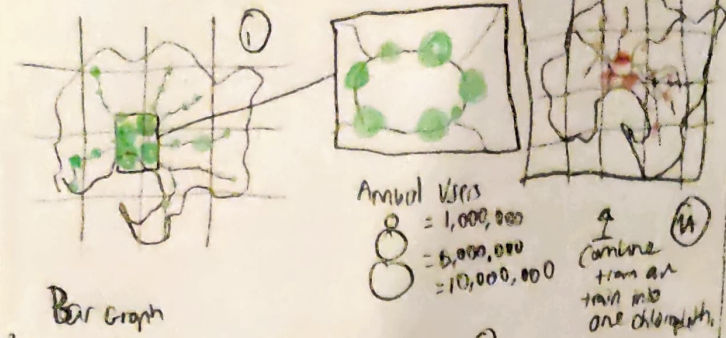
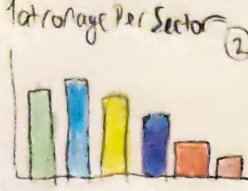


Ideas

Proportional Symbol Map w/ Zoomed-in Minimap



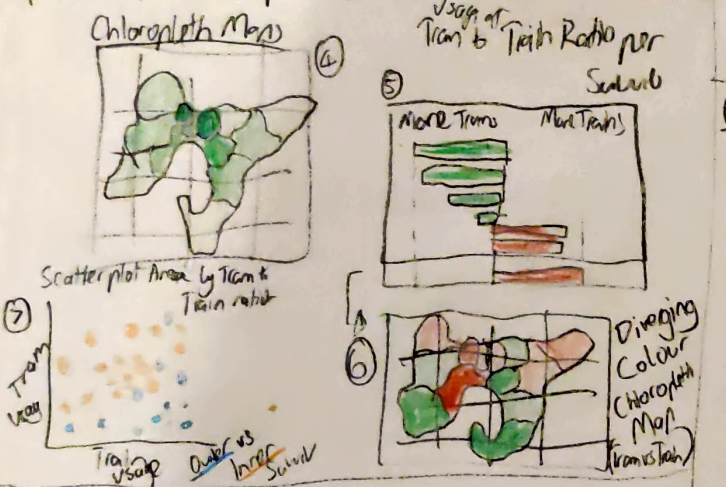
Bar Graph



Pie Chart Percentage Train & Tram Users



Number of Tram Stops per LGA/SAHA • Train Stops per LGA/SAHA



Combine and Refine

Individual choropleth maps between trams and trains could be converted into a diverging colour choropleth depending on the data available.
 eg ④... → ⑥...

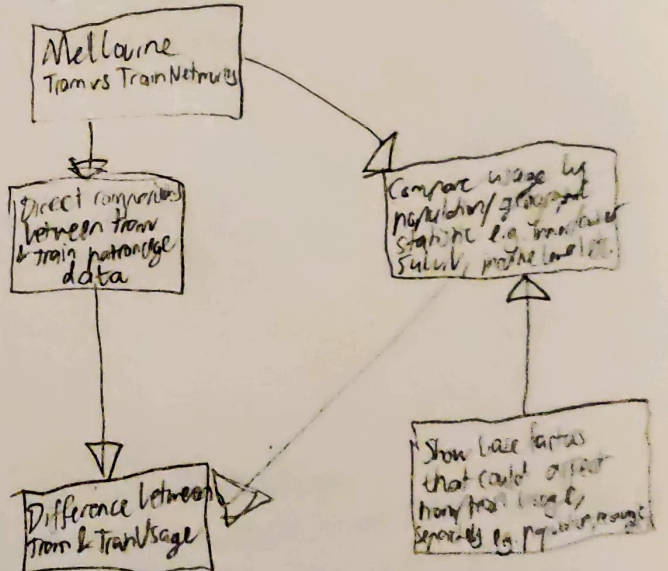
Rather than having individual tram/train lines colored in ②, have them either colored green or red for tram/train and combine each result into one graph to keep the visualisation consistent with its colour theme, and for ease of understanding.

Consider adding interactive elements allowing for one visualisation to be transformed into another without the need for extra space.

Filter

- 1) Likely not a possibility as the tram network stops are very densely located compared to train stops which could make comparison difficult, instead keep each stops (tram/train) in a separate symbol map like ①
- 2) A bar graph of each train/tram line by usage rates may not be possible as some stops serve multiple lines meaning the data could be duplicated. Would need to remove frequently duplicated stops from each sector e.g. City stops from Tram, City Loop from Trains.

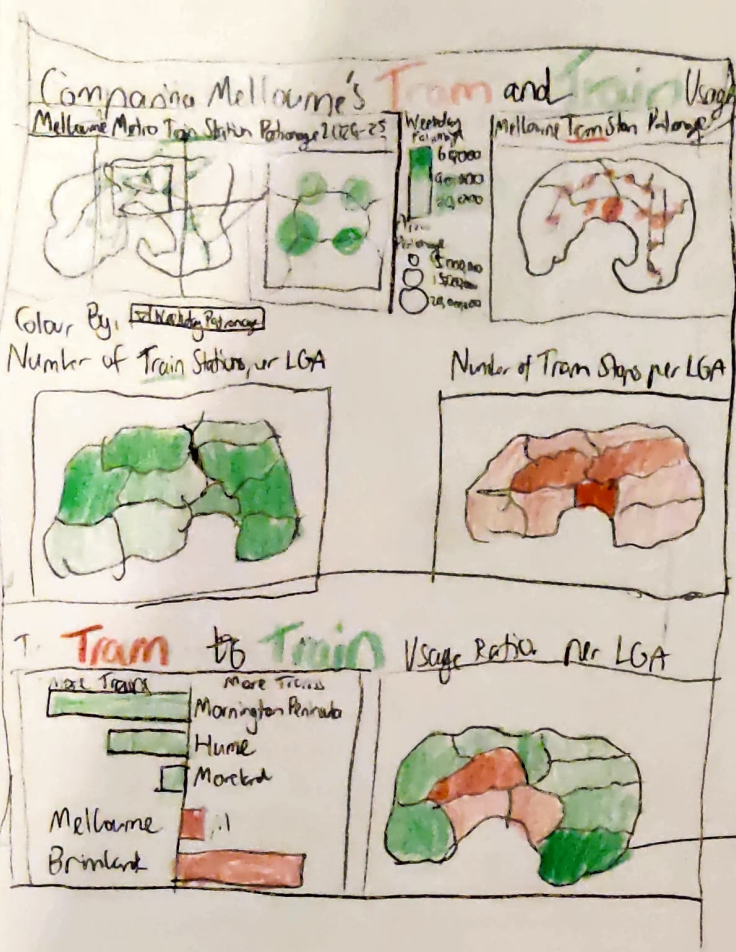
Categorie



Question

- Are there any correlations between tram vs train usage and particular patterns eg - Outer growing suburbs more reliant on trains than inner city.
- Do the visualisations make meaningful comparisons between tram and train patronage/usage etc?
 * Probably, but further analysis may be required e.g. from population data sources.
- Do the visualisations contain enough layers/information to allow for further insights.
 * Consider adding layers for roads.

Layout



Title: Melbourne Tram vs Train Usage

Author: Zachary Gale

Sheet: 2

Task: Portait Page

Operations

User selects data applicable to each visualisation

- Patronage by weekday
- Patronage on Saturday
- Patronage on Sunday
- Patronage on New Year's Day

Hovering over relevant data point could show tooltip with all data points

Station Name: Port Phillip
Annual Patronage: 15,000,000
Weekday Patronage: 4,000,000
Saturday Patronage: 7,000,000
Sunday Patronage: 6,500,000

Focus

- Greater emphasis on comparison between tram and train network usage
- Clear Column-based Structure used to compare tram and train networks with lower half where both are compared directly.
- Some focus on interactivity with the use of filters, tooltips. Capable of using brushing to zoom in on a particular LGA's tram & train usage rates.

One section focusing on train-related data



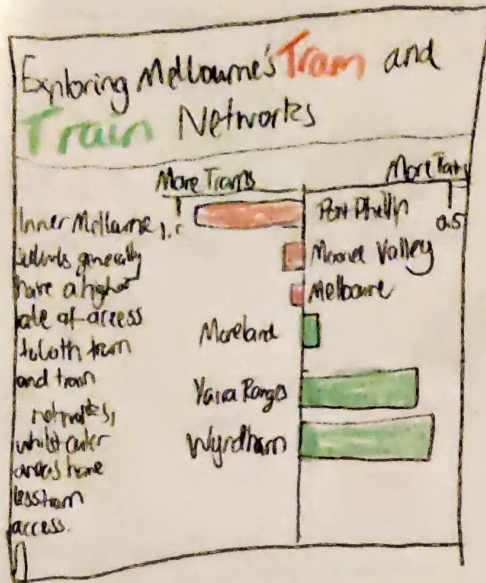
One section focusing on tram related data

A lower section combining the data to compare both trams & trains directly.

Discussion

- Clear Column/Grid based layout
- All sections seem relatively connected even when the data is directly compared in the lower section.
- The bottom two choropleth charts may be too similar, an extra visualisation may need to be added to account for this.
- Brushing may be difficult to implement, especially if a selected LGA is used. Would the first map zoom into the LGA when it is selected in the lower half's maps?
- Likely room to add more visualisations in this layout.
- Legends/keys would need to be clearly separated perhaps the leftmost visualisations could have the keys to the left, and the rightmost to the right.

Layout



Title: Melbourne Tram vs. Train Usage

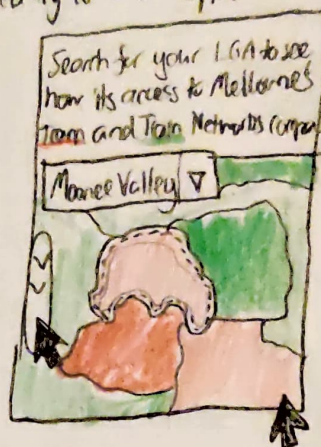
Author: Zachary Gane

Sheet: 5

Task: Long-form/Scrollytelling Article

Operations

User scrolls through web page to reveal new visualisations relating to the topic



LGA: Melbourne

Tram to Train Ratio: 1.1

Tram Usage Annual: x

Train Usage Annual: x

Tooltips can also be hovered over to view data related to specific LGA/suburb.

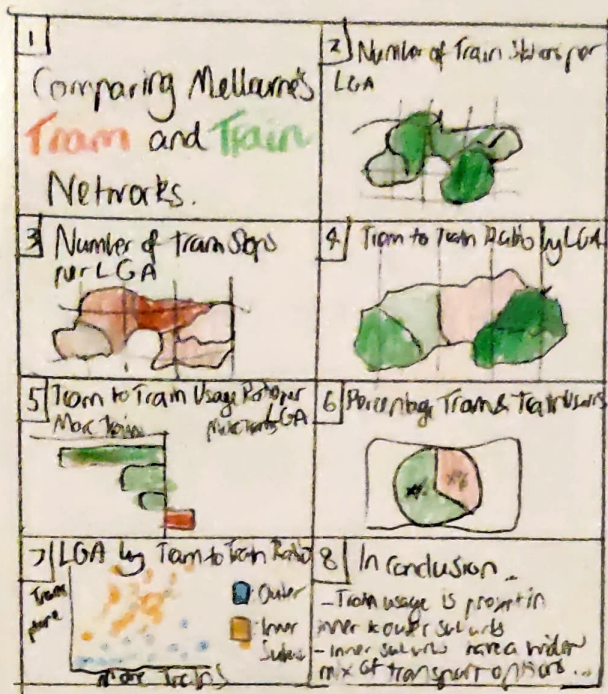
Focus

- Focus is placed on interactivity and prioritise the user's ability to arrive at their own conclusions relating the data being presented.
- Designed to catch user's attention and encourage user to view all aspects of the visualisation, as each idiom is presented separately.
- Greater emphasis on direct comparisons b/w Melbourne's tram & train networks.

Discussion

- Scrollytelling format allows for a limited number of visualisations to be displayed simultaneously.
- Requires greater optimisation to allow the visualisation to run on many devices.
- If the ^{initial} idioms presented are not interesting, the user may not view the other following idioms.
- May not be feasible in the scope of the assignment.
- Data-link ratio is improved as there are ideally seamless transitions between idioms in this format.

Layout



Title:

Author: Zachary Gale

Sheet: 4

Task: Presentation - Style Visualisation

Operations



- Individual slides are presented to data visualisation uses.

- One slide at a time with complementary commentary/explanation from visualisation creator

- Users could potentially view the slides on their own too.

Focus

- Focus placed on communication with visualisation users, the visualisation/slides are a complementary component.

- Less emphasis placed on direct comparison between tram and train stop usage as both can't be present simultaneously unless they are incorporated into one idiom.

Discussion

- Perhaps not best suited for a web page as the visualisation creator would ideally need to present the visualisation uses as well.

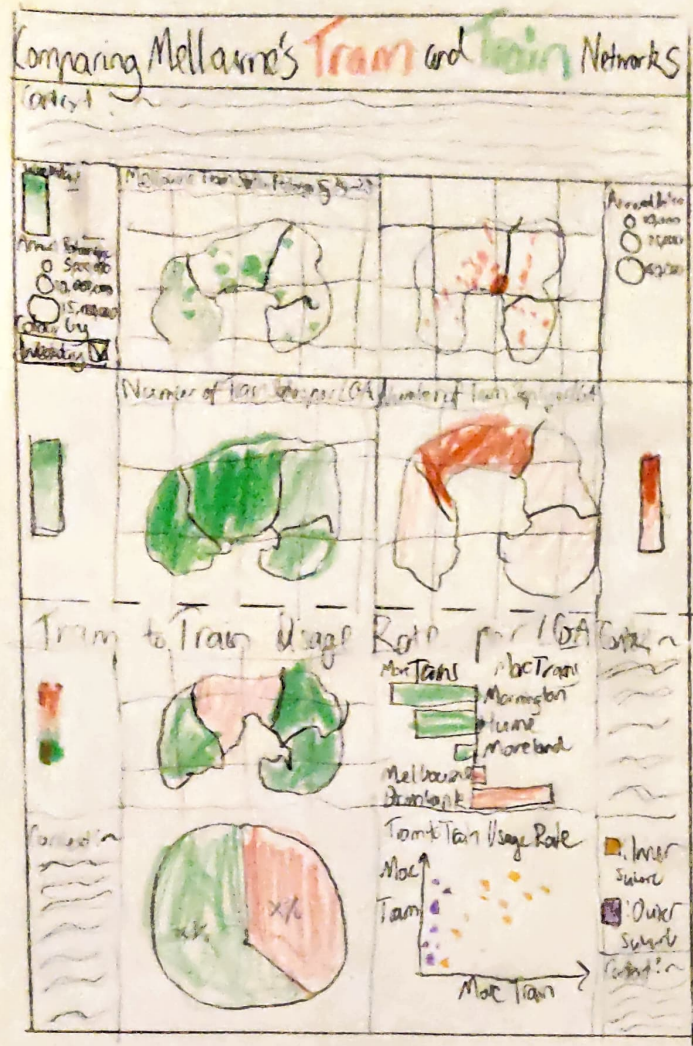
- Operations would be limited as the presenter would be in control of the navigation of the web page rather than the data user.

- Best suited for a professional environment.

- Limited reach, available to lesser users.

- The visualisation may not be comprehensible without the visualisation creator present.

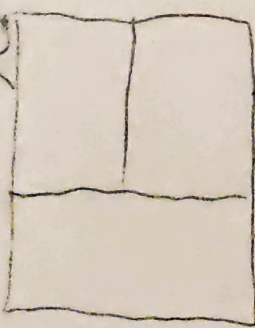
Layout



Focus

- Great emphasis on direct comparisons and combined comparisons between Melbourne's tram and train networks
- Clear column-based / dual-column layout with extra space on leftmost & rightmost sides of visualisation for keys, legends, annotations and/or context paragraphs.
- Some focus on interactivity with the use of filters, dynamic tooltips and the potential for brushing.

One section focusing specifically on train-related data idioms



One section focusing specifically on tram-related data idioms.

A lower section featuring direct/combined comparative idioms between trains & trams or other idioms.

User selects data applicable to each visualisation, changes the color scale for relevant idiom.

☐	Weekday
☐	School Holiday
☐	Saturday
☐	Sunday

Hovering over relevant data points with mouse provides full information for the data point

E.g.

Station Name	Flinders St.
Annual Usage	10,000,000
Weekday Usage	4,000,000
Saturday Usage	700,000
Sunday Usage	6,500,000

LGA: Murrumbidgee
Tram to Train Ratio: 0.17
Number of Trams: 2
Number of Trains: 15

- Potential for brushing, e.g. when user selects an LGA and all other maps and the selected one zooms into LGA

Detail

- Uses the sheet 2 portrait layout and add extra idioms, annotations and text as space.
- Clear dual-column layout with extra space for keys & legends.
- Idioms that are similar are grouped with same title and section.
- Potential to add space for zoomed mini maps.
- Less whitespace between idioms.